



# DAYANANDA SAGAR COLLEGE OF ENGINEERING

(An Autonomous Institute affiliated to Visvesvaraya Technological University (VTU), Belagavi,  
Approved by AICTE and UGC, Accredited by NAAC with 'A' grade)  
Shavige Malleshwara Hills, Kumaraswamy Layout, Bengaluru-560 111, India



## DEPARTMENT OF ELECTRONICS&COMMUNICATION ENGINEERING

(Accredited by NBA Tier 1: 2025-2028)

**COURSE NAME: Interdisciplinary Project Based Learning**

**COURSE CODE: 1BPRJ208**

**Second Semester B.E 2025-2026**

## SMART WATER LEVEL MANGEMENT SYSTEM

*Submitted in partial fulfillment for the award of the degree of*

**Bachelor of Engineering**

**In**

**Electronics and Communication Engineering**

*Submitted by*

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**VISVESVARAYA TECHNOLOGICAL UNIVERSITY  
JNANASANGAMA, BELAGAVI-590018, KARNATAKA, INDIA  
2025-26**

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## DEPARTMENT OF ELECTRONICS&COMMUNICATION ENGINEERING



### CERTIFICATE

Certified that the project report entitled “**SMART WATER LEVEL MANGEMENT SYSTEM**” carried out by **AZHAR ALI BOHRA** bearing USN:**1DS25EC058** and **BOORAVALI SAI PRANAV** bearing USN:**1DS25EC067** are bonafide students of **DAYANANDA SAGAR COLLEGE OF ENGINEERING**, an autonomous institution affiliated to VTU, Belagavi in partial fulfillment for the award of Degree of **Bachelor of Electronics and Communication Engineering** during the year **2025-2026**. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The mini project report has been approved as it satisfies the academic requirements with respect to the work prescribed for the said Degree.

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### External Viva

This project work report is submitted for the Interdisciplinary Project Based Learning Viva-voce examination held on \_\_\_\_\_.

**Name of the Examiners**

1. ....

2. ....

**Signature with date**

.....

.....

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## DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

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## DECLARATION

We, **AZHAR ALI BOHRA (1DS25EC058), BOORAVALI SAI PRANAV (1DS25EC067), DEVASHISH H C (1DS25ME023) and DHRUV GARG (1DS25ME024)**, respectively, hereby declare that the mini project work entitled “ **SMART WATER LEVEL MANGEMENT SYSTEM** ” has been independently done by us under the guidance of ‘**Dr. JAMUNA S’, Professor**, ECE department and ‘ **Dr. NINGEGOWDA B.M.** ’, Associate professor, ME department submitted in partial fulfillment of the requirement for the award of the degree of **Bachelor of Electronics & Communication Engineering** at **Dayananda Sagar College of Engineering**, an autonomous institution affiliated to VTU, Belagavi during the academic year 2025-2026.

We further declare that we have not submitted this report either in part or in full to any other university for the award of any degree.

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# ABSTRACT

The primary objective of this project is to develop a Smart IoT-Based Automatic Water Management and Control System that significantly optimizes water efficiency, completely eliminates manual intervention, and prevents critical resource wastage such as tank overflow and pump dry-running. By tackling these everyday challenges, the project actively supports the United Nations Sustainable Development Goals, specifically SDG 6 (Clean Water and Sanitation), SDG 7 (Affordable and Clean Energy), and SDG 12 (Responsible Consumption and Production). To achieve these robust goals, the proposed methodology utilizes an ESP32 microcontroller as the central processing unit, directly interfaced with physical float switches, a 5V relay module, and an electric water pump. This hardware configuration operates on a continuous, real-time feedback loop; the mechanical float switches monitor the volumetric capacity of the storage tank, prompting the ESP32 to autonomously activate the pump when water levels drop below a minimum threshold and instantly deactivate it the moment maximum capacity is reached. Furthermore, the system capitalizes on the ESP32's built-in Wi-Fi capabilities to introduce advanced IoT functionalities, bridging the physical hardware with a custom mobile application. The main outcomes of this technical integration yield a highly cost-effective and energy-efficient solution that not only automates physical water distribution but also empowers users with live telemetry data, including real-time water usage monitoring, comprehensive pump run-time analytics, and secure remote control capabilities. Consequently, this system proves to be highly practical and applicable for everyday deployment in residential homes, multi-story apartments, and smart small-scale municipal water infrastructures. Looking toward the future scope, the project's modular architecture can be seamlessly expanded to support precision indoor garden irrigation, allowing the framework to incorporate additional environmental sensors to autonomously regulate and dispense the exact amount of water needed for household plants, thereby broadening its overall utility in modern smart home ecosystems.

**Keywords: Internet of Things (IoT), Water Management System, ESP32 Microcontroller, Resource Efficiency ,Automation**

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## LIST OF ABBREVIATIONS

| Abbreviation | Description                        |
|--------------|------------------------------------|
| IOT          | Internet of Things                 |
| IDE          | Integrated development environment |
| LoRaWaN      | Long Range Wide Area Network       |
| GPIO         | General purpose input/output       |
| BJT          | Bipolar Junction Transistor        |
| TDS          | Total dissolved solids             |

# Chapter 1

## INTRODUCTION

Water is an essential resource for daily life, agriculture, and industrial activities. Efficient water management has become increasingly important due to growing population, urbanization, and rising water demand. In many homes, apartments, and small industries, water tanks are still monitored manually, which can lead to water overflow, pump dry running, energy wastage, and inefficient use of water resources. These challenges create the need for an automated and intelligent water management system. The **Smart Water Level Management System Using ESP32** is designed to automate the monitoring and control of water levels in storage tanks. The system uses an ESP32 microcontroller, float switches, a relay module, and a water pump to regulate the water filling process automatically. The water level is monitored, and the pump is switched ON or OFF based on predefined water levels without requiring human intervention. Through IoT connectivity, users can also monitor the system status and control the pump remotely using the Blynk mobile application.

The main objective of this project is to reduce water wastage, minimize manual effort, and improve the efficiency of water usage. The system helps prevent tank overflow, ensures reliable pump operation, and provides real-time monitoring and remote access capabilities. The scope of the project includes residential buildings, educational institutions, farms, and small commercial establishments where efficient water management is required.

The project is significant because it promotes sustainable use of water and energy resources while providing a cost-effective and user-friendly solution. It supports Sustainable Development Goals (SDGs) such as SDG 6, SDG 7, SDG 11, SDG 9 – Industry, Innovation and Infrastructure, and SDG 12 – Responsible Consumption and Production. The expected outcome is a reliable automated water management system capable of monitoring water levels, controlling pump operation, providing real-time updates, and improving overall water conservation and operational efficiency.

## Chapter 2

### LITERATURE SURVEY

The integration of the Internet of Things (IoT) into water management has driven significant advancements in resource conservation, automated distribution, and quality monitoring.

Recent research has explored various sensor networks, communication protocols, and domain-specific applications to address the growing challenges of water scarcity and infrastructure inefficiency.

A major focus in recent literature is the development of real-time monitoring systems for water tanks and distribution networks. Parveen et al. [4] implemented a standard IoT architecture for continuous water level tracking that successfully delivers mobile alert notifications. Building on the need for versatile deployment, Joshika et al. [5] proposed a scalable water distribution model adaptable to both low-tech rural pipelines and complex urban infrastructure. Furthermore, Agarkar et al. [8] designed a framework explicitly targeted at reducing non-revenue water (wastage and leaks) through real-time flow monitoring, while Divya et al. [2] utilized a combination of sensors and cloud connectivity for real-time tank monitoring and automated pump control.

Despite these advancements, several limitations persist. A critical vulnerability is the high dependency on stable Wi-Fi or internet connections, with many systems lacking robust offline fallback control mechanisms [2]. Additionally, foundational IoT tracking systems often fail to adequately address data security and potential cyber vulnerabilities in the communication layer [4]. Finally, highly advanced flow-monitoring frameworks often face friction during deployment, as they are difficult to integrate seamlessly with existing legacy mechanical water infrastructures without requiring total replacement [8].

To overcome internet dependency and power constraints, researchers have evaluated alternative communication protocols. For localized environments, K. S et al. [3] developed a cost-effective water level controller using Arduino and Bluetooth, allowing users to toggle pumps via smartphones. For long-range requirements, particularly in villages with more facilities, Manoharan and Rathinasabapathy [6] employed LoRaWAN for the low-power transmission of water quality and consumption data over vast distances.

Bluetooth-based solutions are fundamentally constrained by proximity, entirely lacking global remote monitoring capabilities [3]. Conversely, while LoRaWAN effectively solves the range and power issues, current research focuses too heavily on communication hardware and lacks the integration of Artificial Intelligence (AI) for predictive consumption analysis [6].

Smart water concepts have also been highly tailored to specific domains. In agricultural settings, Durai et al. [1] successfully automated irrigation using IoT soil moisture sensors to maximize crop yield and reduce water waste. In the context of sanitation, Pu [7] designed an IoT-based system to monitor purification stages (such as pH and turbidity) in real-time, ensuring treated water meets strict safety standards.

These domain-specific solutions often suffer from isolation and resource intensity. Agricultural systems typically focus strictly on farm parameters and lack integration with broader community or urban water supply networks [1]. Meanwhile, the continuous multi-sensor sampling required in water treatment facilities leads to high power consumption and currently lacks predictive maintenance features for treatment filters [7].

The existing literature demonstrates the viability of IoT for smart water management but highlights a clear need for more resilient architectures. Future systems must address the gaps in offline fallback capabilities, bridge the divide between localized and long-range monitoring, prioritize data security, and ensure backward compatibility with legacy infrastructure while optimizing power consumption.

## Chapter 3

# SYSTEM DESIGN & METHODOLOGY

### 3.1 System Architecture

The Smart IoT-Based Automatic Water Management and Control System is designed to automate water tank monitoring and pump control using the ESP32 microcontroller. The system consists of two float switches, a relay module, a water pump, LED indicator, and the Blynk IoT platform. The ESP32 continuously monitors the status of the low level and high level float switches installed inside the water tank. Based on the water level detected by these switches, the controller automatically turns the pump ON or OFF through a relay module.

The ESP32 is connected to a Wi-Fi network, enabling communication with the Blynk cloud server. Users can monitor tank status and pump operation remotely through the Blynk mobile application. The LED provide local visual indications of tank conditions.

### 3.2 Block Diagram

Figure 3.1 illustrates the interaction between the sensing, processing, control, and monitoring units of the system.

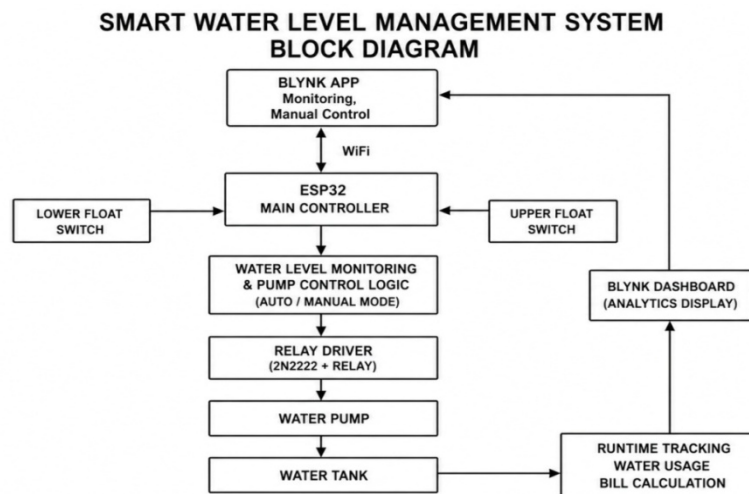


Figure 3.2(Block Diagram)

### 3.3 Hardware Components

#### ESP32 Microcontroller-

The ESP32 serves as the central processing unit of the system. It reads water level information from the float switches, executes control logic, connects to Wi-Fi, and communicates with the Blynk platform.

#### Float Switches- Two float switches are used

- Low-Level Float Switch – Detects when the water level falls below the minimum level.
- High-Level Float Switch – Detects when the tank reaches its maximum level.

These switches provide digital inputs to the ESP32.

Relay Module-The relay module acts as an electrically isolated switch that enables the ESP32 to control the water pump safely.

Water Pump-The pump transfers water into the storage tank whenever the tank level falls below the predefined minimum level.

LED Indicator-The LED provides a visual indication when the tank becomes full.

Power Supply-A DC adapter provides power to the ESP32 and relay module while supplying sufficient voltage to operate the water pump.

Transistor- 2N2222-Used in Relay driver circuit

Resistors- used as current limiting resistors

Jumper Wires- Used for connectivity

BreadBoard- for prototyping

### 3.4 Software Components

Arduino IDE-The Arduino IDE is used to write, compile, and upload the program to the ESP32.

Blynk IoT Platform-Blynk provides a cloud-based interface for remote monitoring and control.

Users can view:

- Tank Status
- Pump Status
- Water Level Condition
- System Alerts

ESP32 Firmware-The firmware continuously reads sensor inputs, executes control logic, updates Blynk values, and controls the relay output.

### 3.5 Circuit Diagram

The circuit is designed using ESP32 GPIO pins connected to float switches, relay module, LED.

Connections-

| Component | ESP32 Pin |
|-----------|-----------|
|-----------|-----------|

|                  |         |
|------------------|---------|
| Low Float Switch | GPIO 26 |
|------------------|---------|

|                   |         |
|-------------------|---------|
| High Float Switch | GPIO 27 |
|-------------------|---------|

|              |         |
|--------------|---------|
| Relay Module | GPIO 25 |
|--------------|---------|

|     |         |
|-----|---------|
| LED | GPIO 33 |
|-----|---------|

Power Connections:

- ESP32 powered through USB/DC supply.

-DC motor pump powered through 12v dc adapter

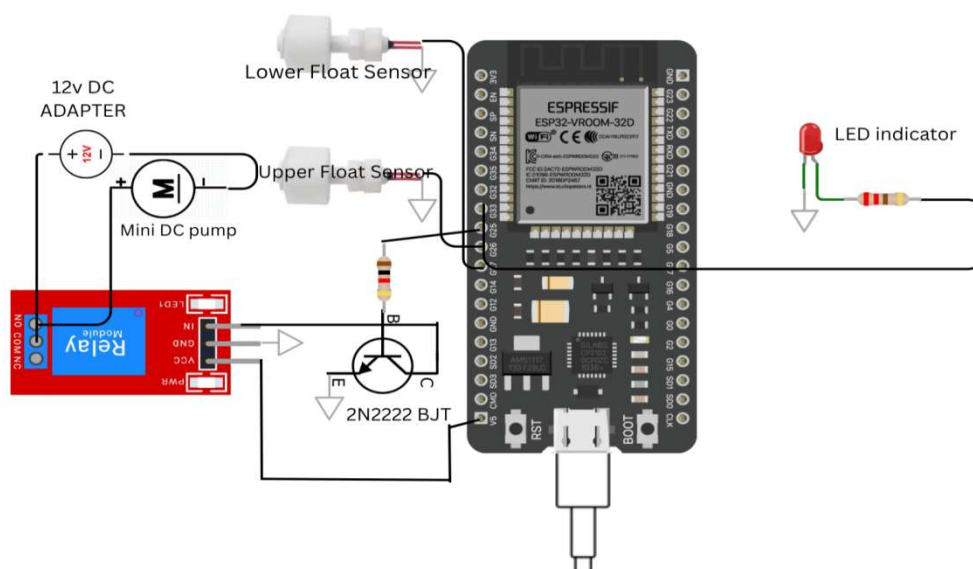


Figure 3.3 Circuit Diagram

### 3.6 Flowchart

Figure 3.3 follows the basic logic of the circuit and demonstrates the working of the circuit logic

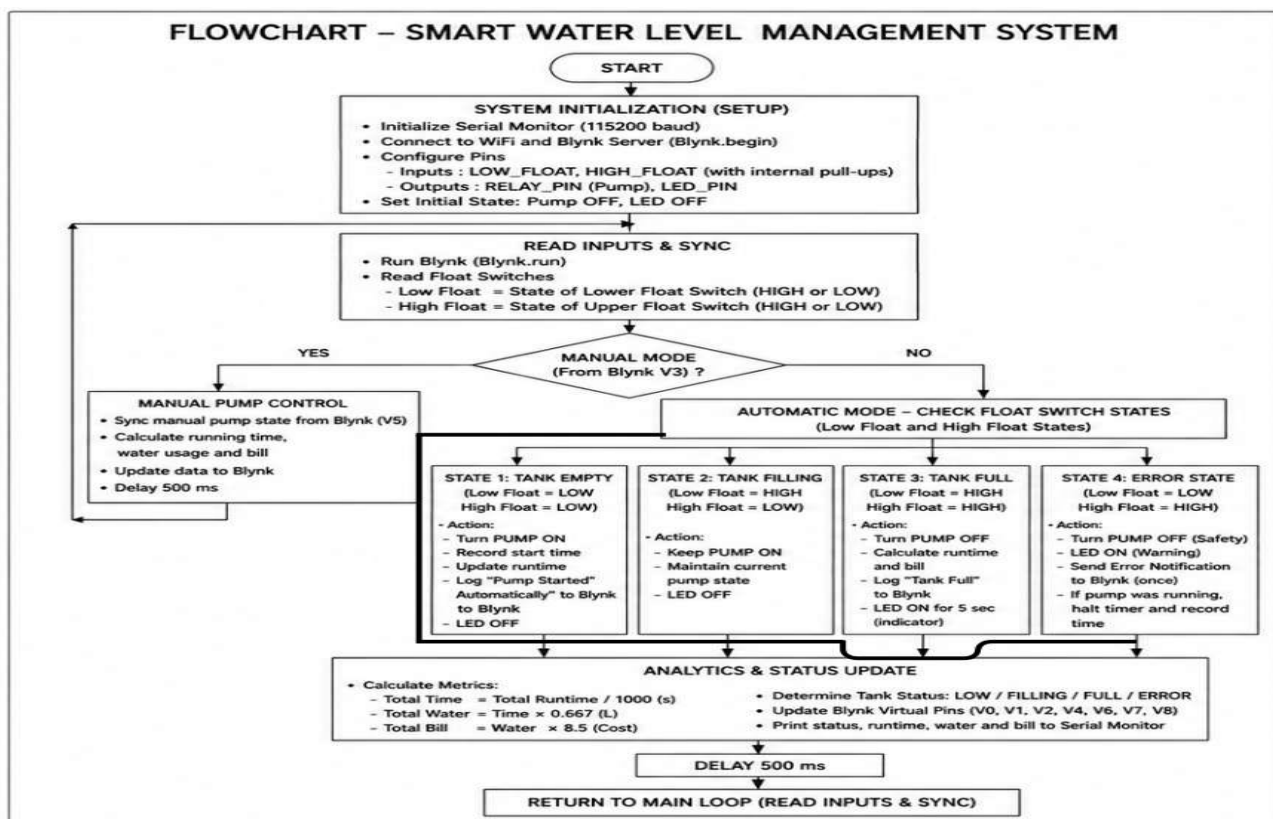


Figure 3. 4(Flowchart)

### **3.7 Algorithm**

Step 1: Start the system.

Step 2: Initialize ESP32 GPIO pins and Blynk connection.

Step 3: Connect ESP32 to Wi-Fi network.

Step 4: Read the status of the low-level float switch.

Step 5: Read the status of the high-level float switch.

Step 6: If the low-level float switch indicates low water level, turn ON the pump.

Step 7: Keep the pump running while the high-level float switch remains LOW.

Step 8: When the high-level float switch becomes HIGH, turn OFF the pump.

Step 9: Turn ON the LED to indicate tank full condition.

Step 10: Send updated tank and pump status to the Blynk application.

Step 11: Repeat the monitoring cycle continuously.

### **3.8 Working Methodology**

The system operates automatically based on the water level inside the storage tank. When the water level drops below the low-level float switch, the ESP32 detects the condition and activates the relay module, turning ON the water pump. The pump continues to fill the tank while the controller monitors both float switches.

Once the water level reaches the high-level float switch, the ESP32 deactivates the relay, turning OFF the pump. At the same time, the LED indicator is activated to indicate that the tank has reached full capacity. The system status, including pump condition and tank level information, is transmitted to the Blynk cloud platform via Wi-Fi. Users can monitor the system remotely using the Blynk mobile application.

The automation eliminates the need for manual monitoring, prevents water overflow, reduces water wastage, and improves the overall efficiency of water management. The combination of IoT connectivity and automatic control makes the system suitable for residential, educational, agricultural, and small industrial applications.

## Chapter 4

### SNAPSHOTS AND RESULTS

#### 4.1 Snapshots

Following Figure 4.1 shows the main circuit of the project. The circuit includes ESP32 microcontroller with relay to switch motor on/off with a blinking led light as an indicator . It includes 2nnn transistor to drive the relay,current limiting resistors and floats wired upto GPIO pins of the ESP32

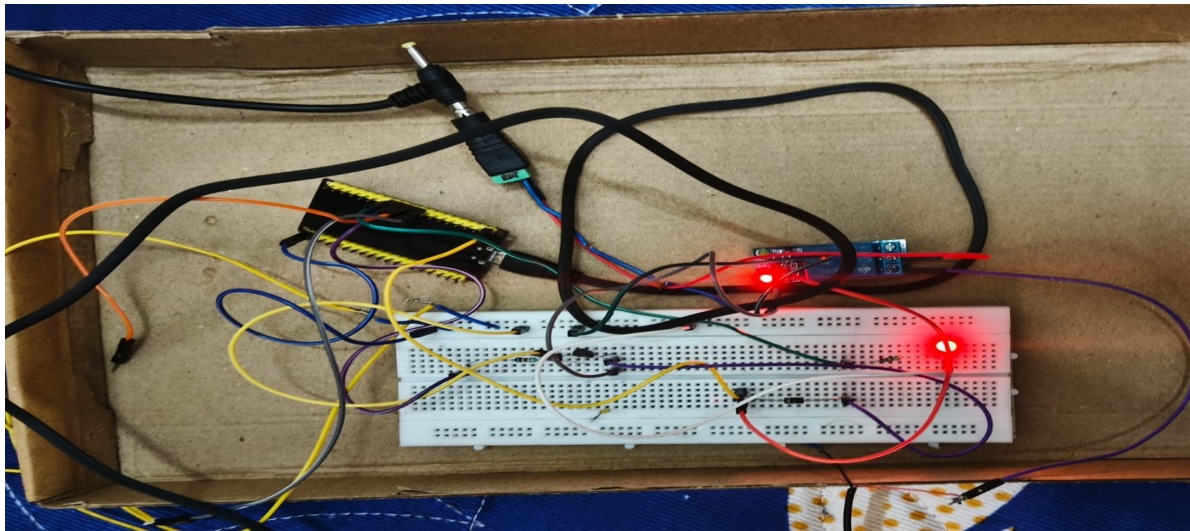


Figure 4.1 Working circuit

Figure 4.2 demonstrates blynk app alert notification when the pump reaches full state

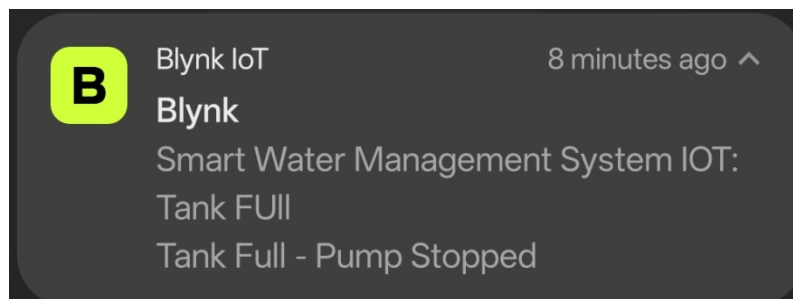


Figure 4.2 Mobile app notification

Figure 4.3 shows the Blynk web app that is indicating all the parameters of the system such as high and low float logic, pump status, pump run time, tank status, water used as well as the real time bill of the water used according to bwssb tariffs. Mode switch button is also provided.

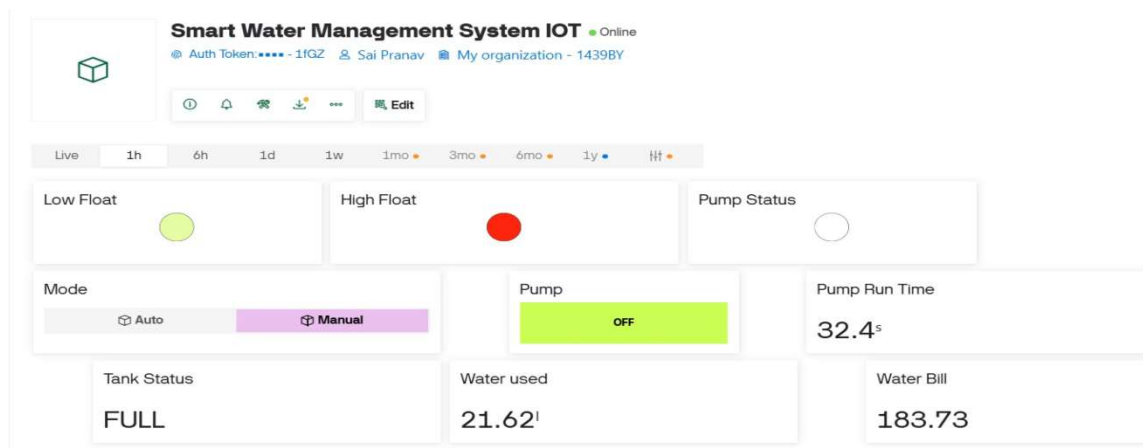


Figure 4.3 Blynk interface

Figure 4.4 shows blynk mobile app that shows the same features and statistics as the webapp allowing to monitor water usage remotely

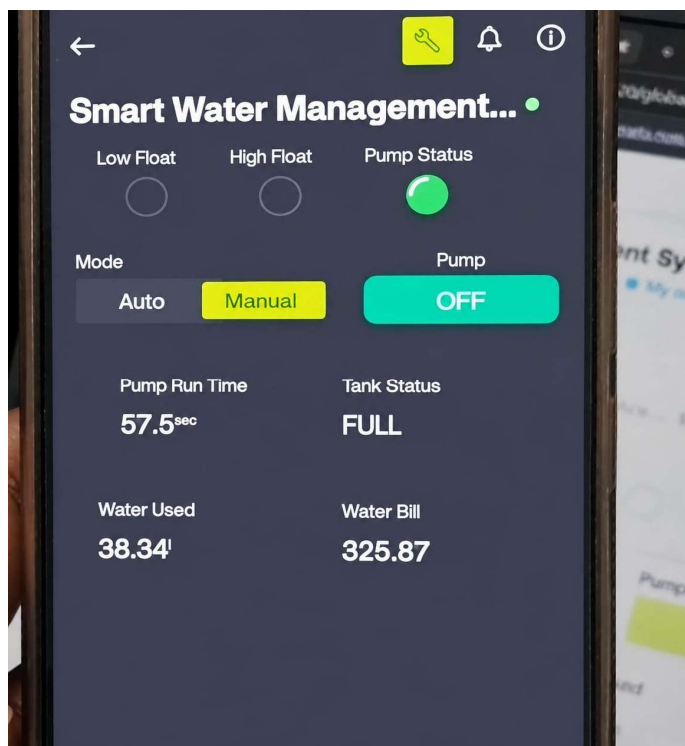


Figure 4.4 mobile app interface

## 4.2 Results

Table 4.1 demonstrates the difference between conventional systems and Proposed IoT system

| Feature                | Conventional System | Proposed IoT System |
|------------------------|---------------------|---------------------|
| Water Level Monitoring | Manual              | Automatic           |
| Pump Operation         | Manual              | Automatic & Remote  |
| Overflow Prevention    | Not Reliable        | Reliable            |
| Water Usage Tracking   | Not Available       | Available           |
| Water Bill Estimation  | Not Available       | Available           |
| Mobile Monitoring      | Not Available       | Available           |
| Remote Pump Control    | Not Available       | Available           |

Table 4.1 (Comparison table)

## Chapter 5

# CONCLUSION AND FUTURE ENHANCEMENTS

### 5.1 Conclusion

The Smart IoT-Based Automatic Water Management and Control System using ESP32 successfully automates water level monitoring and pump control. By utilizing float switches, a relay module, and the Blynk IoT platform, the system efficiently manages water levels without continuous human intervention. Features such as automatic pump operation, remote monitoring, runtime analytics, tank-full alerts, and error detection help reduce water wastage, save energy, and improve overall system reliability. The project provides a cost-effective and user-friendly solution suitable for residential, educational, agricultural, and small-scale industrial applications.

### 5.2 Future Enhancements

The system can be further improved through:

- Integration with advanced cloud platforms for data storage and analysis.
- Mobile notifications through SMS, email, and push alerts.
- AI-based prediction of water consumption and pump operation.
- Ultrasonic sensors for more accurate water level measurement.
- Water quality monitoring using pH, turbidity, and TDS sensors.
- Leakage detection using flow and pressure sensors.
- Solar-powered operation for improved energy efficiency.
- Multi-tank monitoring and control through a single dashboard.

These enhancements can make the system more intelligent, efficient, and suitable for future smart water management applications.

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## APPENDIX

Code link- <https://reentry.co/smartwaterlevelmanagementsystem>